Segment Based Movie Promotion and Content Discovery Strategy

MovieLens 10M end to end decision support project

Field	Details
Course	Big Data Analytics Final Project 2026
Dataset	MovieLens 10M from GroupLens Research
Technology	MySQL 8.0, KNIME Analytics Platform and Power BI
Models	K Means audience segmentation and Logistic Regression
Student name and ID	_____
Submission date	16 October 2026

Academic Integrity Declaration

By signing below, I certify that the attached assignment is the project team's own work. All datasets, publications, software and AI-assisted work are acknowledged. The team remains responsible for the database execution, KNIME workflows, dashboard, analysis, interpretations and recommendations.

Signature _____ Date _____

AI Use Statement

Generative AI assisted with planning, technical explanation, code and workflow review, report drafting, formatting and chart preparation. All numerical claims were checked against repository outputs and official documentation. AI-generated text was not treated as independent evidence.

Abstract

This project develops an end-to-end decision-support solution for segment-specific movie promotion using MovieLens 10M. MySQL stores 10,000,054 ratings and derived data products. KNIME reads directly from MySQL, performs reproducible preparation, creates four K Means audience segments and trains a Logistic Regression model for positive ratings, then writes segments, predictions and metrics back to MySQL. The classifier achieved 69.26% accuracy and 75.83% ROC-AUC on 76,633 eligible TEST rows, compared with a 59.32% majority baseline. A five-page Power BI dashboard links these results to controlled promotion, de-prioritisation and testing decisions. The conclusions apply to warm-start rating behaviour, not viewing, revenue or causal campaign impact.

Keywords *MovieLens, audience segmentation, K Means, Logistic Regression, MySQL, KNIME, Power BI, content strategy*

1 Introduction and Problem Definition

A movie platform must decide which titles and genres to surface to different users. A single popularity ranking treats all users alike and can repeatedly promote familiar titles even when they fit a segment poorly. The decision-maker in this project is a Content Programming and Personalisation Manager responsible for selecting titles for discovery rows and targeted campaigns.

The objective is to combine behavioural audience segments with positive-rating probabilities so that the decision-maker can identify titles to promote, maintain, reduce or test. A positive rating is defined as a score of at least 4.0. It is a preference proxy; it is not evidence of a view, purchase, subscription renewal or revenue outcome.

Decision Questions

Decision question	Measure	Dashboard evidence
How do rating behaviour and genre preferences differ across users?	Segment size, activity, positive rate and genre affinity	Audience Segments
Which user segments are sufficiently interpretable and large enough to support curation?	K Means profile, segment size and silhouette	Audience Segments and Model Governance
How accurately can prior behaviour identify a positive rating?	Accuracy, precision, recall, F1, ROC-AUC and confusion matrix	Positive Engagement Model
Which lower-popularity titles merit a controlled promotion test?	Predicted and actual positive rate, TRAIN popularity and TEST support	Content Opportunity Matrix

The required pipeline is implemented as Dataset to MySQL to KNIME to model to MySQL result write-back to interactive dashboard to decision recommendations. Each decision question maps to a stored data product and a dashboard page, which prevents the report from presenting model scores without a decision use.

The project contributes a reproducible analytical workflow, two interpretable models, five dashboard pages and four evidence-based recommendations. Its main limitation is applicability: the Logistic Regression feature view covers only 5.11% of chronological TEST rows because both the user and movie require TRAIN history.

2 Dataset Description

MovieLens 10M is a public research dataset released by GroupLens Research in January 2009. The README permits research use subject to its stated conditions and identifies GroupLens as the source and owner of the collection [1]. User identifiers are anonymised, and the dataset contains no names, contact details or demographic attributes.

Entity	Rows	Key fields	Meaning and analytical role
ratings	10,000,054	user_id, movie_id, rating, timestamp	Explicit preference event on a 0.5 to 5.0 scale
movies	10,681	movie_id, title, genres	Movie catalogue and pipe-separated genre labels
tags	95,580	user_id, movie_id, tag, timestamp	Free-text annotations supplied by users
movie_genres	21,564	movie_id, genre	Normalised bridge used for filtering and aggregation
rating users	69,878	user_id	Users represented in the ratings table

The GroupLens README reports 71,567 total users because it also includes users represented only in the tags file. The project database contains 69,878 users with at least one rating. Every rating user has at least 20 ratings. The rating timestamps cover 9 January 1995 to 5 January 2009.

The dataset is relevant because it combines a large interaction table, reusable movie dimensions, multi-value genres and timestamps suitable for future-facing validation. It exceeds the recommended project size by several orders of magnitude and exercises the required database, KNIME and dashboard architecture. However, it does not contain impressions, watch time, licensing cost, revenue or contemporaneous market conditions. Recommendations therefore concern rating preference and controlled discovery tests.

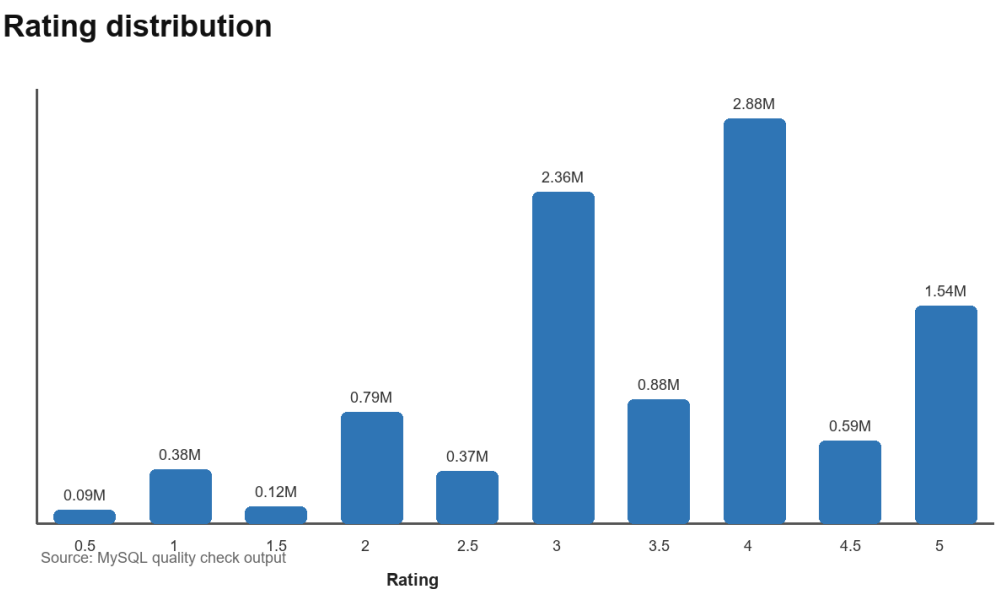


Figure 1 Distribution of all 10,000,054 MovieLens ratings

3 Data Quality Assessment

The project checks completeness, uniqueness, validity, consistency and plausible coverage in both SQL and the reproducible KNIME data-quality workflow. The checks verify referential integrity, allowed rating values, timestamp validity, duplicate user-movie pairs, minimum activity and catalogue coverage. The workflow preserves the original rows and records derived transformations separately.

Quality check	Before cleaning	After cleaning	Treatment and rationale
Rating rows	10,000,054	10,000,054	No deletion or imputation was required
Missing or orphan ratings	0	0	Foreign-key consistency confirmed
Missing or orphan tags	0	0	Foreign-key consistency confirmed
Invalid ratings or timestamps	0	0	Allowed range and positive timestamp rules passed
Duplicate user-movie ratings	No rows returned	No rows returned	No duplicate removal was required
Users below 20 ratings	0	0	Minimum activity condition already satisfied
Catalogue movies with ratings	10,677 of 10,681	10,677 of 10,681	Four unrated titles retained as valid catalogue records

Because the checks found no invalid interaction rows, before and after counts are identical. Cleaning consisted of reproducible structural transformations rather than destructive row removal: parsing genres into movie_genres, assigning chronological splits, creating TRAIN-only user and movie histories, generating user-genre profiles and materialising dashboard marts. Low-support titles remain in storage and are controlled through minimum-support filters at decision time.

The remaining quality risks are conceptual. Ratings are observed only after a user chooses to rate, so missing ratings do not mean dislike. The population overrepresents active raters, and the collection ends in 2009. Genre labels can be broad or multi-valued, while tags are sparse relative to ratings. These limitations affect generalisation even though the stored rows pass integrity checks.

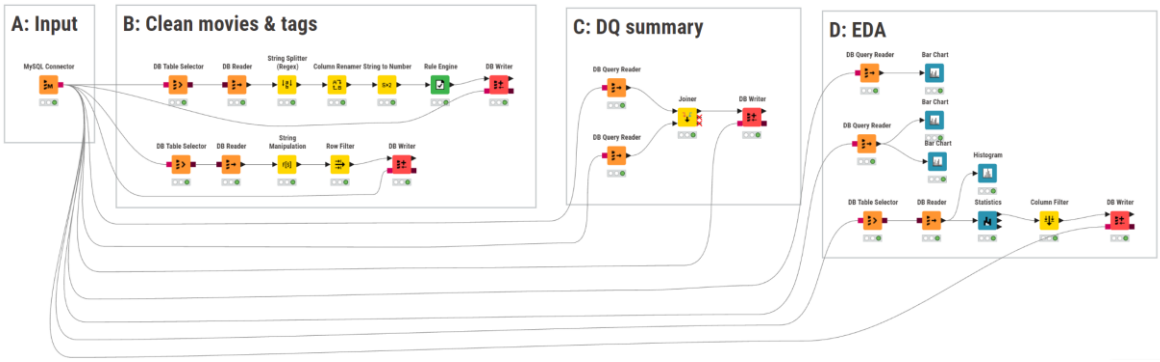


Figure 2 KNIME data quality workflow reading directly from MySQL

4 System Architecture and Data Pipeline

MySQL is the governed storage layer for source records, derived features and model results. KNIME reads directly from MySQL, performs data preparation and modelling, and writes cluster labels, predictions and metrics back to database tables. Power BI reads compact MySQL views instead of importing the full ratings table. This design preserves a traceable handoff between each layer and keeps the dashboard responsive.

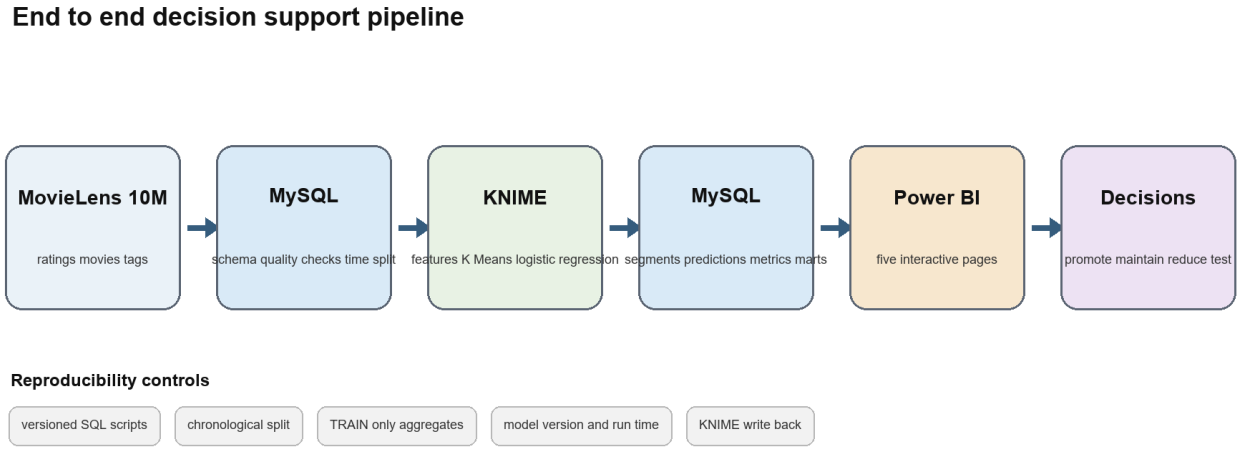


Figure 3 Implemented architecture from source data to decisions

4.1 MySQL Database and SQL

The schema separates source entities from derived and reporting tables. Primary keys protect movie and interaction identity; indexes on user, movie and timestamp support joins, filtering and chronological aggregation. Runnable scripts create the schema, load the delimited files, execute quality checks, create the 70/15/15 split, build TRAIN statistics, assemble Logistic Regression features and publish dashboard views. Filtering, GROUP BY aggregation and joins produce the report KPIs and data-quality evidence.

Layer	Examples	Purpose
Source	ratings, movies, tags, movie_genres	Preserve the documented MovieLens entities
Derived	rating_split, train_user_stats, train_movie_stats, ml_rating_features	Create reproducible chronological and behavioural features
Model results	user_segments, rating_predictions, model_metrics	Store KNIME-written labels, probabilities and evaluation
Reporting	dashboard views	Expose compact and version-aware evidence to Power BI

4.2 KNIME Processing and Result Write Back

Three KNIME workflows cover data quality, audience segmentation and positive-engagement prediction. Database Connector and Reader nodes retrieve governed tables. Labelled workflow sections apply type handling, transformations, validation and modelling. Database Writer operations persist the final user segments, rating-level predictions and model metrics to MySQL. This write-back is mandatory for the project and avoids a CSV-only analytical path.



Figure 4 KNIME Logistic Regression workflow with chronological branches and MySQL write back

Chronological Split and Leakage Controls

Split	Rows	Period	Use
TRAIN	7,000,042	1995-01-09 to 2005-04-02	Feature histories and model fitting
VALID	1,500,006	2005-04-02 to 2006-12-28	Model and threshold checks
TEST	1,500,006	2006-12-28 to 2009-01-05	Final out-of-time evaluation

All user, movie and genre histories are calculated from TRAIN. TRAIN model rows use leave-one-out aggregates so a rating cannot enter its own predictors. VALID and TEST use frozen TRAIN aggregates. user_id and movie_id remain identifiers rather than numerical predictors. The eligible feature view requires both the user and movie to have TRAIN history, which protects feature meaning but reduces TEST coverage.

5 Exploratory Data Analysis

The average rating is 3.512, and 50.06% of ratings meet the positive threshold of 4.0 or higher. Rating 4.0 is the most common value with 2,875,850 records. These values indicate that the binary target is not severely imbalanced at the full-dataset level, although temporal and warm-start filtering changes the composition of the model sample.

KPI	Value	Interpretation
Ratings	10,000,054	Large interaction base for aggregation and temporal validation
Rating users	69,878	Users with explicit rating history
Movies	10,681	Catalogue size available for content analysis
Average rating	3.512	Overall preference level
Positive rating rate	50.06%	Share of ratings at least 4.0
Observation window	1995-01-09 to 2009-01-05	Historical period and temporal boundary

Drama has the largest genre rating volume at 4,344,198 and a 56.52% positive rate. Film-Noir has a 71.95% positive rate but much lower support. This contrast shows why a high positive rate alone is not a sufficient promotion rule. Decisions must consider segment fit, historical popularity and the number of eligible TEST observations.

User activity is highly uneven. Selective Critics later emerge as the most active segment, averaging 385.3 TRAIN ratings per user, while the other final segments average approximately 61 to 67. This difference affects how much title-level support each segment contributes to the content opportunity matrix.

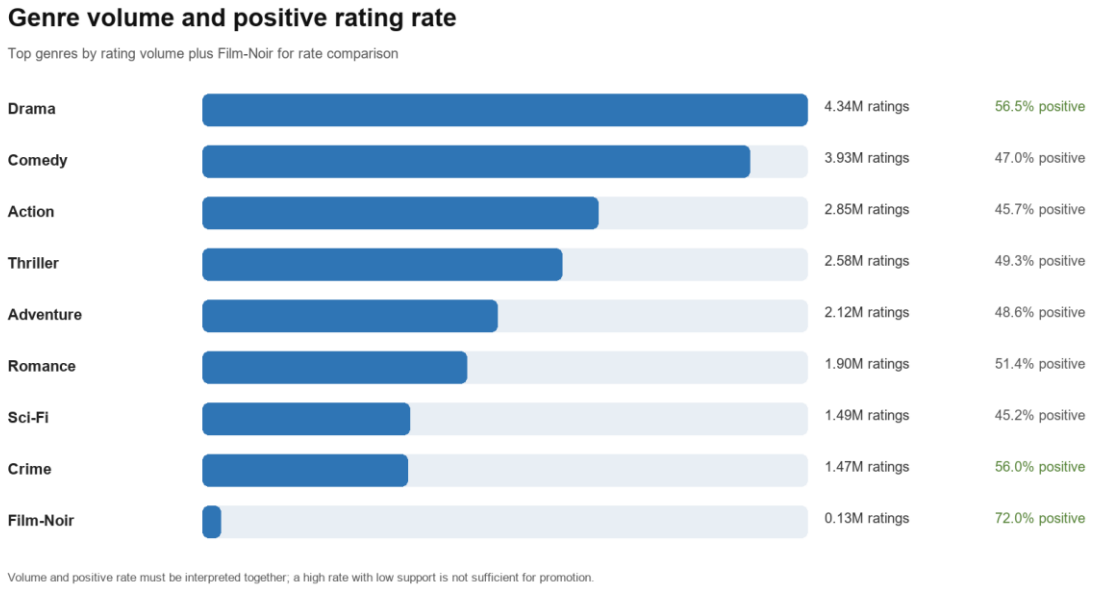


Figure 5 Genre rating volume and positive-rating rate

6 Machine Learning

6.1 K Means Audience Segmentation

K Means answers the decision question about interpretable audience groups. The KNIME workflow joins TRAIN user statistics with pivoted genre averages, converts genre ratings to user-relative affinity features, handles missing values and numerical outliers, standardises the feature set and evaluates K values from 3 through 7. The final cluster labels are written to user_segments with a model version and run date.

K	Mean silhouette	Decision
3	0.060209	Fewer profiles but weaker separation
4	0.068639	Selected for interpretability
5	0.069188	Only 0.00055 above K equals 4
6	0.064601	Lower score
7	0.065112	Lower score

K equals 4 was retained because K equals 5 improved mean silhouette by only 0.00055 while adding another business profile. The absolute silhouette scores remain low, so the segments overlap. They should be treated as descriptive decision lenses rather than natural or demographic categories.

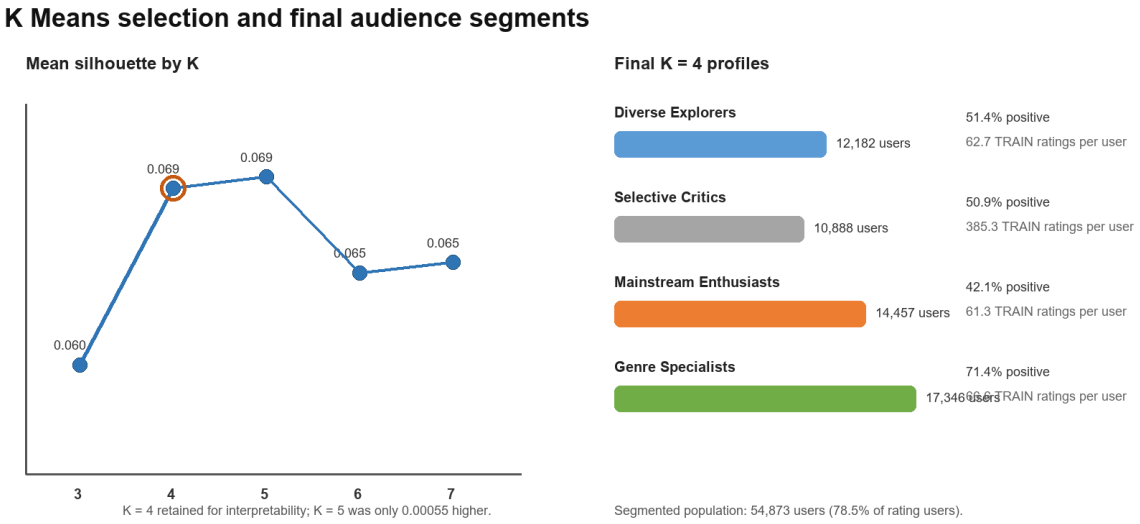


Figure 6 K selection and the four final audience profiles

Segment	Users	Positive rate	Profile interpretation
Diverse Explorers	12,182	51.38%	Relative thriller, action and mystery affinity
Selective Critics	10,888	50.88%	Highest activity and longest observed history
Mainstream Enthusiasts	14,457	42.06%	Relative family and animation affinity
Genre Specialists	17,346	71.43%	Largest segment and highest positive tendency

6.2 Logistic Regression for Positive Engagement

Logistic Regression estimates the probability that a rating is at least 4.0. Predictors describe prior user behaviour, prior movie behaviour, activity, cluster membership and genre context derived from TRAIN. KNIME fits missing-value and standardisation preprocessing on TRAIN and applies the frozen transformations to VALID and TEST. A 0.5 threshold converts probability into class, while SQL recomputes final metrics from the stored TEST predictions.

Metric	Result	Interpretation
Accuracy	69.26%	9.93 percentage points above the majority baseline
Majority baseline	59.32%	Reference accuracy for the dominant TEST class
Precision	60.58%	Share of predicted positives that are positive
Recall	69.94%	Share of actual positives identified
F1	64.92%	Balance between precision and recall
ROC-AUC	75.83%	Ability to rank positive outcomes above negative outcomes

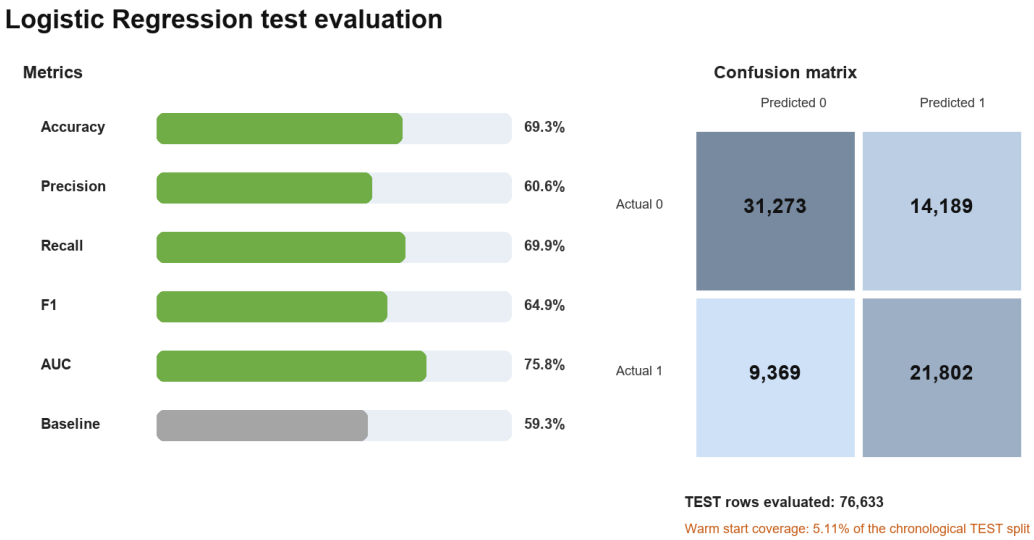


Figure 7 Logistic Regression TEST metrics and confusion matrix

The confusion matrix contains 31,273 true negatives, 14,189 false positives, 9,369 false negatives and 21,802 true positives. The model improves on the baseline and provides useful ranking information for covered rows, but it is not universally applicable. Only 76,633 of 1,500,006 TEST rows have complete warm-start features, giving 5.11% coverage. The reported metrics therefore describe this eligible subset, not the entire future interaction population.

Potential leakage is controlled through chronological splitting, TRAIN-only aggregates, leave-one-out TRAIN features and frozen transformations. Remaining risks include temporal drift, unobserved exposure, selection bias among active raters, overlapping clusters and untested probability calibration.

7 Dashboard and Insights

The Power BI project imports compact MySQL views that include source KPIs and KNIME-written segments, predictions and metrics. Five pages separate descriptive analysis, model interpretation, opportunity selection and governance. Global controls for model version, segment, genre and minimum support prevent incompatible runs or weak cells from being combined.

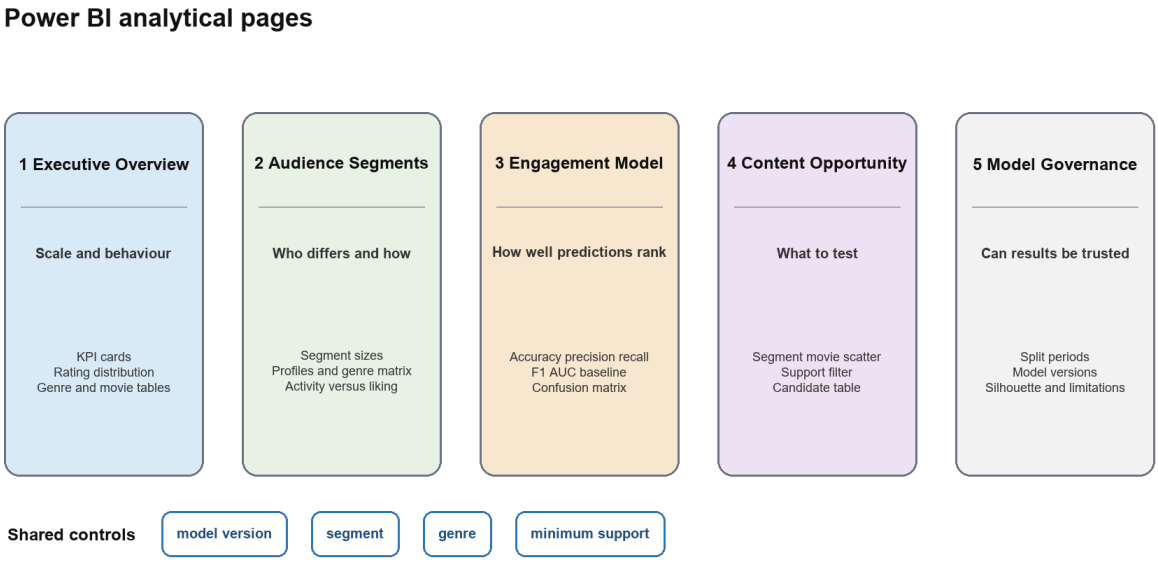


Figure 8 Implemented five-page Power BI report structure from the PBIP definitions

Dashboard page	Question answered	Key visuals and controls
Executive Overview	What is the scale and overall rating behaviour?	KPI cards, rating distribution, genre KPIs and top movies
Audience Segments	How do the four groups differ?	Segment size, activity versus positive rate, profiles and genre heatmap
Positive Engagement Model	How well does the classifier perform?	Baseline comparison, metrics, coefficients and confusion matrix
Content Opportunity Matrix	Which titles should be promoted, reduced or tested?	Popularity versus predicted appeal, candidate table and support filter
Model Governance	When are results safe to interpret?	Split dates, versions, silhouette, coverage and limitations

The dashboard answers the first two decision questions through segment profiles and K selection, the third through out-of-time classification metrics, and the fourth through the opportunity matrix. Only Selective Critics produced title-segment cells with TEST support of at least 50. The other segments had maximum support between 18 and 30, so their title-level results are exploratory rather than action-ready.

The opportunity matrix places TRAIN popularity on a logarithmic x-axis, predicted positive probability on the y-axis and TEST support in bubble size. Median reference lines define four action regions. A single model-version filter is required because combining prediction runs would double counts and invalidate support.

8 Decision Recommendations

The recommendations below answer the decision questions while keeping the action within the available evidence. Each action specifies its empirical basis, expected benefit and principal trade-off.

Recommended action	Supporting evidence	Expected benefit and trade off
Run a Selective Critics discovery test led by Hotel Rwanda and My Neighbor Totoro	Hotel Rwanda: support 108, predicted 76.72%, actual 77.78%, TRAIN popularity 215. My Neighbor Totoro: support 58, predicted 74.87%, actual 74.14%, popularity 487.	Increase discovery of lower-popularity titles. A controlled experiment is required because rating evidence does not prove campaign lift.
Add City of God and Oldboy to the same controlled candidate pool	City of God: support 60, predicted 71.66%, actual 70.00%, popularity 968. Oldboy: support 52, predicted 69.94%, actual 67.31%, popularity 72.	Broaden the discovery slate with aligned predicted and observed preference. The evidence applies only to eligible warm-start users.
Reduce default Selective Critics placement for Home Alone, Star Wars Episode I, Twister and The Mask	Each pair has support above 50 but low actual positive rates: 17.65%, 19.61%, 10.17% and 5.66%, respectively, despite high historical popularity.	Reduce segment-specific mismatch while keeping the titles available elsewhere. Monitor diversity and engagement loss.
Deploy segment-aware ranking only when warm-start evidence is available	Classifier TEST coverage is 5.11%; segment write-back covers 54,873 of 69,878 rating users, or 78.5%.	Avoid unsupported predictions. Use popularity or editorial rules as fallback and track coverage as a deployment KPI.

Evaluation Plan

Eligible users should be randomly assigned to the existing experience or the proposed segment-specific slate. Placement, time window and exposure count should remain consistent. Primary outcomes should include click-through, play-start, completion and post-view positive rating when those fields become available. Confidence intervals and minimum sample sizes should be reported before expanding the treatment.

The Incredibles should be treated separately as a test case rather than an automatic promotion. Its predicted positive rate is 74.45% but its observed rate is 60.75% across 107 qualifying ratings. The disagreement provides a useful diagnostic for model calibration, title context or temporal change.

9 Conclusion and Limitations

The project implements the required end-to-end decision-support architecture. A documented public dataset is loaded into MySQL; KNIME reads the database, validates and transforms the data, fits K Means and Logistic Regression models, and writes results back; Power BI consumes MySQL views; and recommendations are tied to model evidence, observed response, popularity and support.

Four audience profiles provide interpretable curation lenses, although their low silhouette values show substantial overlap. The Logistic Regression model improves accuracy by 9.93 percentage points over the majority baseline and achieves 75.83% ROC-AUC on eligible TEST rows. The strongest immediate application is a controlled discovery test for Selective Critics, where title-level support is sufficient to compare predicted and observed positive response.

Limitations

- MovieLens 10M ends in January 2009 and may not reflect current catalogue, platform or audience behaviour.
- Ratings reveal expressed preference after a rating event, not exposure, viewing, revenue, churn or causal campaign impact.
- The minimum of 20 ratings per rating user overrepresents active users and limits cold-start relevance.
- The Logistic Regression model covers only 5.11% of chronological TEST rows because user and movie histories must exist in TRAIN.
- K Means silhouette scores are low in absolute terms, and segment names are interpretive summaries rather than demographic identities.
- High-support title-level evidence is concentrated in Selective Critics; other segment recommendations require more observations.

Potential Improvements

Future work should add cold-start features, probability calibration, ranking metrics, stronger rolling temporal validation and exposure-aware outcomes. Modern viewing, impression, completion, licensing-cost and revenue data would support decisions beyond stated preference. Online experiments should estimate causal effects before any broad promotion policy is adopted.

The final decision is therefore bounded: use the current outputs to prioritise transparent, measurable experiments for warm-start users, while keeping editorial or popularity fallbacks for uncovered cases.

References

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Appendix A Reproducibility Guide

Credentials must be stored in the ignored python_sql_scripts .env file or in KNIME credential configuration. No password should be committed to the repository.

- Download MovieLens 10M from GroupLens and verify the source files.
- Run python_sql_scripts 01 through 05 to create the schema, load source data, execute quality checks, create chronological splits and build TRAIN feature tables.
- Open knime/movielens_audience_segmentation.knwf, configure the MySQL Connector, evaluate K values, retain K equals 4 and write user_segments.
- Run python_sql_scripts/06_logistic_features.py to create ml_rating_features.
- Open knime/movielens_like_prediction.knwf, execute TRAIN, VALID and TEST, and write rating_predictions and model_metrics.
- Run python_sql_scripts/07_dashboard_views.py and refresh dashboard/movielens_content_strategy.pbix or the PBIP project.
- Confirm split dates, row counts, model version and final metrics against this report before submission.

Required deliverable	Repository artifact
SQL schema, import and queries	sql/01_create_schema.sql through sql/07_dashboard_views.sql
KNIME workflow export	knime/movielens_data_quality_finalterm.knwf, movielens_audience_segmentation.knwf and movielens_like_prediction.knwf
Interactive dashboard	dashboard/movielens_content_strategy.pbix and PBIP project
Setup and dependencies	README.md plus python_sql_scripts/README.md and requirements.txt
Final report	output/MovieLens_Content_Strategy_Final_Report.docx; export the approved copy to PDF

Appendix B Technical Evidence



Figure B1 KNIME audience segmentation workflow

Result table	Producer	Core fields	Dashboard use
user_segments	KNIME K Means	user_id, cluster, segment label, model version	Segment profiles and filters
rating_predictions	KNIME Logistic Regression	probability, predicted class, actual class, split, version	Metrics and opportunity matrix
model_metrics	KNIME and SQL	metric, value, split, run metadata	Model performance and governance

The report figures are derived from repository outputs, including the quality-check workbook, K Means CSV results, stored Logistic Regression predictions, MySQL dashboard views and PBIP page definitions. The source artifacts remain the authoritative evidence for reproduction.